VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 10 | PAPER - 2

JEE 2022

Date	Maximam Marks	Duration
13th March, 2022	183	3 Hours

General Instructions

1. The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**).

   Each Part has **Three** sections (Section 1, Section 2 & Section 3).

3. **Section 1** contains **7 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

4. **Section 2** contains **7 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

5. **Section 3** contains **2 Paragraphs containing 2 Questions** on each paragraph**.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

6. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

Candidate's Signature : ____________ Invigilator's Signature ____________

MARKING SCHEME

SECTION - 1

- This section contains **SEVEN (07)** Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If only (all) the correct option(s) is(are) chosen
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–1 In all other cases.

SECTION - 2

- This section consists of **SEVEN (07)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–2 In all other cases.

SECTION - 3

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph**.** Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	–1 In all other cases.

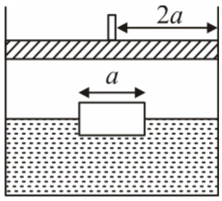
SUBJECT I : PHYSICS — 61 MARKS

SECTION 1

SINGLE CORRECT ANSWERS TYPE

This section contains 07 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. A cubical block of side a is floating in a fixed and closed cylindrical container of radius $2a$ kept on the ground. Density of the block is ρ, whereas density of the liquid is 2ρ. Container is made up of conducting wall, so that the temperature remains constant. A piston is mounted in the cylinder which can move inside the cylinder without friction. If piston oscillates with a large amplitude A, then



(A) the cube will remain stationary
(B) the cube will oscillate with very small amplitude in opposite phase with piston
(C) the cube will oscillate with very small amplitude in same phase with piston
(D) the cube will oscillate with amplitude A

2. Two ammeters, 1 and 2, have different internal resistances: r_1 (known) and r_2 (unknown). Each ammeter has an analog scale such that the angular deviation of the needle from zero is proportional to the current. Initially, the ammeters are connected in series and then to a source of constant voltage. The deviations of the needles of the ammeters are θ_1 and θ_2, respectively. The ammeters are then connected in parallel and then to the same voltage source. This time, the deviations of the needles are θ_1' and θ_2' respectively. Find r_2.

(A) $r_2 = r_1 \dfrac{\theta_1\theta_2}{\theta_1'\theta_2'}$ **(B)** $r_2 = r_1 \dfrac{\theta_1\theta_1'}{\theta_2\theta_2'}$ **(C)** $r_2 = r_1 \dfrac{\theta_2\theta_1'}{\theta_1\theta_2'}$ **(D)** $r_2 = r_1 \dfrac{\theta_1\theta_2'}{\theta_2\theta_1'}$

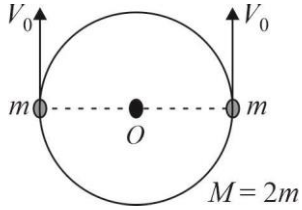
3. The stopping potential for the photo electrons emitted from a metal surface of work function 1.7eV is 10.4V. Identify the energy levels corresponding to the transitions in hydrogen atom which will result in emission of wavelength equal to that of incident radiation for the above photoelectric effect.

(A) $n = 3$ to 1 **(B)** $n = 3$ to 2 **(C)** $n = 2$ to 1 **(D)** $n = 4$ to 1

4. A Young's double slit experiment is performed with monochromatic light of wavelength λ. The slit separation is d. Let P and Q be two completely dark points on the screen on either side of the central maximum with angular position θ. The number of points of maximum between P and Q is:

(A) $\dfrac{d}{2\lambda}\sin\theta$ **(B)** $\dfrac{d}{\lambda}\sin\theta$ **(C)** $\dfrac{2d}{\lambda}\sin\theta$ **(D)** $\dfrac{4d}{\lambda}\sin\theta$

5. In a gravity free space, a smooth ring of mass $M = 2m$ with two small beads each of mass m is as shown in the figure. Initially both the beads are at diametrically opposite points and are given velocity v_0 each in same direction. The speed of the beads just before they collide for the first time is :



(A) v_0 **(B)** $\dfrac{2}{\sqrt{3}}v_0$ **(C)** $\dfrac{v_0}{2}$ **(D)** $\dfrac{\sqrt{3}}{2}v_0$

6. An artificial satellite is first taken to a height equal to half the radius of Earth, let E_1 be the energy required. It is then given the appropriate orbital speed such that it goes in a circular orbit at that height. Let E_2 be the energy required. The ratio $\frac{E_1}{E_2}$ is :

(A) 4 : 1 **(B)** 3 : 1 **(C)** 1 : 1 **(D)** 1 : 2

7. The length of a cylinder is measured with a metre rod having least count 0.1 cm. Its diameter is measured with Vernier calipers having least count 0.01cm. Given that length is 5.0 cm and radius is 2.0 cm. The percentage error in the calculated value of the volume will be :

(A) 1% **(B)** 2% **(C)** 2.5% **(D)** 4%

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 07 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

8. A charge particle having a charge q experience a force $\vec{F} = q\left(-\hat{j}+\hat{k}\right)N$ in a magnetic field $\vec{B}$ when it has velocity equal to $1\hat{i}\ m/s.$ The force becomes $\vec{F} = q\left(\hat{i}-\hat{k}\right)N$ when the velocity is changed to $\vec{v}_2 = 1\hat{j}\ m/s.$ Choose the correct options:

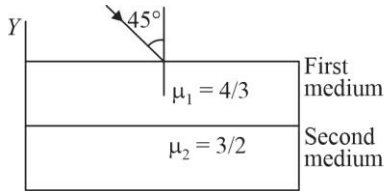
(A) $\vec{B} = (\hat{i} - \hat{j} + \hat{k})T$

(B) $\vec{B} = (\hat{i} + \hat{j} + \hat{k})T$

(C) Charge projected with velocity $-2\hat{i} + \hat{j} + \hat{k}$ m/s from origin moves on a circle with radius $\sqrt{2}m\ [q = 1C,\ m = 1kg]$

(D) Charge moving along line $\frac{2x-3}{2} = y - 1 = z$ will experience a zero force

9. Two thin slabs of refractive indices $\frac{4}{3}$ and $\frac{3}{2}$ are placed parallel to each other in XY-plane as shown in the figure A ray of light in XY-plane is incident from air at an angle of incidence 45°. Choose the correct options.



(A) The unit vector along ray in first medium is $\frac{3}{4\sqrt{2}}\hat{i} - \frac{\sqrt{23}}{4\sqrt{2}}\hat{j}$

(B) The unit vector along ray in second medium is $\frac{2}{3\sqrt{2}}\hat{i} - \frac{\sqrt{14}}{3\sqrt{2}}\hat{j}$

(C) The unit vector along ray in first medium is $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

(D) Speed of light is more in second medium than first medium

10. Two point charge q and –q are separated by distance *l*. Charges are kept on x axis at equal distance from origin. Choose the correct statement(s).

(A) Electric flux through circle $y^2+z^2=R^2,\ x=0$ is $\frac{q}{\epsilon_0}\left[1-\frac{\ell}{\sqrt{4R^2+\ell^2}}\right]$

(B) Electric flux through *y*-*z* plane is $\frac{q}{\epsilon_0}$

(C) Electric flux through the sphere $x^2+y^2+z^2=\ell^2$ is zero

(D) If $E(x)$ represents x-component of electric field at a point (x, y, z) then $E(x)$ is maximum at origin

11. A particle initially at rest at the origin starts moving at $t = 0$ with acceleration $a(t)=Pt\,\hat{i}+Q\,\hat{j}$, where P and Q are positive constants and t is time. The particle passes through the point (a, a) at $t=T$. Which of these options is/are correct ?

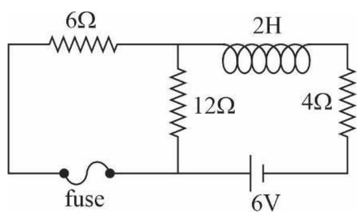
(A) $T=\left(\frac{6a}{P}\right)^{1/3}$

(B) $T=\left(\frac{2a}{Q}\right)^{1/2}$

(C) At $t = T$, the angle made by the velocity of the particle with the positive X-axis is $\tan^{-1}\left(\frac{2}{3}\right)$

(D) At $t = T$, the angle made by the velocity of the particle with the positive X-axis is $\tan^{-1}\left(\frac{1}{3}\right)$

12. For the circuit shown in the figure, which of the following statements is/are correct? [Neglect any resistance of fuse].



(A) Its time constant is 0.25 second

(B) In steady sate, current through inductance will be equal to zero

(C) In steady state, current through the battery will be equal to 0.75 amp

(D) In steady state, suddenly the fuse blows, emf induced across the inductor just after the fuse blows is 6V.

13. A spherical star of radius 10^9 m with surface temperature 6000 K emits electromagnetic radiation like a black body. A small spherical body is situated at a distance 10^{11} m from the star. Which of these options is/are correct? (Stefan-Boltzmann constant $=5.67\times10^{-8}$ W/m^{2}K^4)

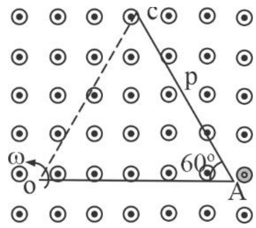
(A) The intensity of radiation reaching the spherical body is close to 13.5 kW/m^2

(B) The intensity of radiation reaching the spherical body is close to 7.4 kW/m^2

(C) The temperature of the spherical body is close to 423 K

(D) The temperature of the spherical body is close to 366 K

14. A metallic V shaped rod OAC is rotated with respect to one of its end in uniform magnetic field such that axis of rotation is parallel to the direction of magnetic field. Length of each arm of rod is L and angle between the arms is 60°. P is the mid-point of section AC. Magnitude of magnetic field is B. Then choose the correct relations.



(A) $V_A - V_0 = \dfrac{\omega BL^2}{2}$ **(B)** $V_A - V_C = \dfrac{\omega BL^2}{2}$

(C) $V_C - V_P = \dfrac{\omega BL^2}{8}$ **(D)** $V_A - V_P = \dfrac{\omega BL^2}{8}$

SECTION 3

This section contains 2 Paragraphs containing 2 Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16

A long coaxial cable consists of a solid inner cylindrical conductor of radius R_1 and a thin outer cylindrical shell of radius R_2. At one end the two conductors are connected together by a resistor and at the other end they are connected to a battery. Hence, there is a current i in the conductors and a potential difference V between them. Neglect the resistance of the cable itself.

15. The electric field E in the region $R_2 > r > R_1$, *i.e.*, between the conductors is:

(A) $\dfrac{2V}{r\ln\left(\dfrac{R_2}{R_1}\right)}$ **(B)** $\dfrac{V}{r\ln\left(\dfrac{R_2}{R_1}\right)}$ **(C)** $\dfrac{V}{2r\ln\left(\dfrac{R_2}{R_1}\right)}$ **(D)** $\dfrac{4V}{r\ln\left(\dfrac{R_2}{R_1}\right)}$

16. Find the magnetic energy in the region between the conductors per unit length and electrical energy per unit length.

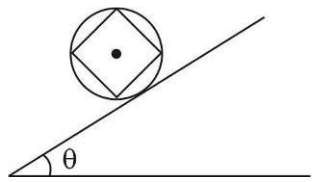
(A) Magnetic energy $= \dfrac{\mu_0 i^2}{2\pi}\ln\left(\dfrac{R_2}{R_1}\right)$, Electrical energy $= \dfrac{\pi\varepsilon_0 V^2}{4\ln\left(\dfrac{R_2}{R_1}\right)}$

(B) Magnetic energy $= \dfrac{\mu_0 i^2}{\pi}\ln\left(\dfrac{R_2}{R_1}\right)$, Electrical energy $= \dfrac{\pi\varepsilon_0 V^2}{\ln\left(\dfrac{R_2}{R_1}\right)}$

(C) Magnetic energy $= \dfrac{\mu_0 i^2}{4\pi}\ln\left(\dfrac{R_2}{R_1}\right)$, Electrical energy $= \dfrac{\pi\varepsilon_0 V^2}{2\ln\left(\dfrac{R_2}{R_1}\right)}$

(D) Magnetic energy $= = \dfrac{\mu_0 i^2}{4\pi}\ln\left(\dfrac{R_2}{R_1}\right)$, Electrical energy $= \dfrac{\pi\varepsilon_0 V^2}{\ln(R_2/R_1)}$

Paragraph for Q-17 to Q-18



Four identical rods of mass M = 6kg each are welded at their ends to form a square and then welded to a massive ring having mass m = 4kg having radius R = 1m. If the system is allowed to roll down the incline of inclination $\theta = 30°$, determine :

17. The moment of inertia of the system about the centre of ring will be :

(A) $20\,kg\ m^2$ **(B)** $40\,kg\ m^2$ **(C)** $10\,kg\ m^2$ **(D)** $60\,kg\ m^2$

18. The minimum value of friction coefficient to prevent slipping is :

(A) $\frac{5}{7}$ **(B)** $\frac{5}{12\sqrt{3}}$ **(C)** $\frac{5\sqrt{3}}{7}$ **(D)** $\frac{7}{5\sqrt{3}}$

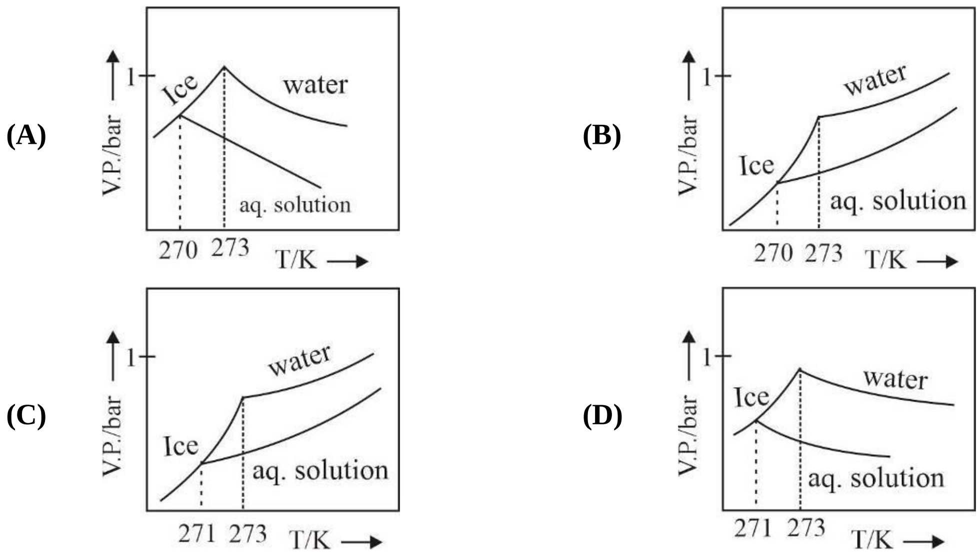
SUBJECT II : CHEMISTRY 61 MARKS

SECTION 1

SINGLE CORRECT ANSWERS TYPE

This section contains 07 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Pure water freezes at 273K and 1 bar and boils at 373K and 1 bar. Addition of non-volatile solute changes the freezing and boiling point of water. Boiling point of an aqueous solution is 100.56°C Latent heat of fusion and latent heat of vaporisation of water are 80 $Cal.g^{-1}$ and 540 $Cal.g^{-1}$ respectively. The figures shown below represent plots of vapour pressure (V.P) versus temperature (T). Identify the correct plot.



2. For the following concentration cell, $Ag(s)|AgCl(s)|Cl^-(aq)\,|\,Ag^+(aq)\,|\,Ag(s)$ if concentration of Ag^+ in anode half cell is $1/10^{th}$ times of concentration of Ag^+ in cathode half cell. The value of $\Delta G(KJ/mole)$ for the cell is :

(A) – 5.7 **(B)** 11.4 **(C)** 5.7 **(D)** – 11.4

3. The conversion of graphite to diamond reduces its volume by X ml/mole isothermally at T = 298K. The pressure at which graphite is in equilibrium with diamond is 1.45×10^4 bar. The standard Gibbs free energy of formation of diamond at $T = 298K$ is $2.9 KJ/mole$. Then value of X is :

(A) 4 **(B)** 6 **(C)** 2 **(D)** 8

4. Which of the following combination will produce nitric oxide gas ?

(A) Zn metal with conc. HNO_3 **(B)** Cu metal with conc. HNO_3

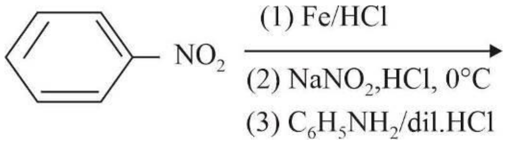
(C) Cu metal with dil. HNO_3 **(D)** Zn metal with dil. HNO_3

5. The oxidation state of the sulphur atom in Marshall's acid, Pyrosulphurous acid, Hyposulphurous acid and Thiosulphurous acid respectively :

(A) + 3, + 4, + 6, + 1 **(B)** + 6, + 1, + 3, + 4

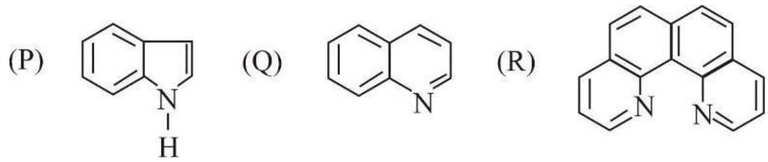
(C) + 6, + 4, + 1, + 3 **(D)** + 6, + 4, + 3, + 1

6. Colour of the major product (p) of the following reaction is :



(A) Orange **(B)** Yellow **(C)** Violet **(D)** Green

7. The order of basicity among the following compounds is :



(A) Q > R > P (B) R > Q > P **(C)** R > P > Q **(D)** Q > P > R

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 07 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

8. Which of the following statement(s) is (are) correct about surface properties:

(A) Adsorption always leads to a decrease in enthalpy and increase in entropy of the system

(B) Adsorption decreases the surface energy

(C) The easily liquefiable gases adsorb more on solid

(D) Greater the surface area per unit mass of the adsorbent, the greater is its capacity of adsorption

9. Hydrazine $(N_2H_4)(g)$ was enclosed in a sealed vessel at certain temperature T_1. At this temperature it undergo following decomposition reaction.

$$3N_2H_4(g) \rightleftharpoons 4NH_3(g) + N_2(g)$$

If temperature is increased from T_1 and T_2 then :

(A) Partial pressure of NH_3 will increase

(B) Partial pressure of N_2H_4 will decrease

(C) Removal of N_2 will shift reaction in forward direction

(D) Simultaneous decomposition of NH_3 will shift equilibrium in reserves direction

10. According to Arrhenius equation the correct option(s) among the following is(are) :

(A) When $T \to \infty; \kappa = A,$ here A is pre-exponential factor

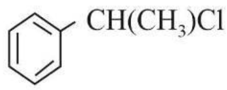
(B) For an exothermic reaction, rate of forward reaction increases with temperature

(C) The reaction with higher E_a value is more temperature sensitive

(D) Rate constant increases with increase in temperature

11. For the following compound, the correct statement(s) with respect to nucleophilic substitution reactions is (are) :

$C_2H_5COCH_2Cl$ (i)



(ii)

$C_2H_5OCH_2Cl$ (iii)

(iv)

(A) I and II follow S_N2 mechanism

(B) III and IV follow S_N1 mechanism

(C) The order of reactivity for I, II and III for S_N2 reaction is I > II > III

(D) Transition state of compound IV is most stable

12. Among the following, the correct statement(s) is(are) :

(A) All carbon atoms of $-CH_3$ groups do not lie in the same plane in $Al_2(CH_3)_6$

(B) $AlCl_3$ can act as lewis acid in both monomer and dimer form

(C) Bridging H_b-B-H_b bond is stronger than terminal $B-H_t$ bond in diborane

(D) Maximum six hydrogen atoms lie in the same plane in diborane

13. The options(s) with only amphoteric oxide is(are) :

(A) Bi_2O_3, N_2O_5, GeO_2, SiO_2 **(B)** SnO, PbO, N_2O, N_2O_5

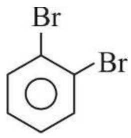
(C) BeO, SnO, SnO_2, Ga_2O_3 **(D)** $A\ell_2O_3$, SnO_2, PbO_2, Ga_2O_3

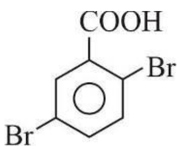
14. A compound A, $C_{13}H_8ONBr_3$ on hydrolysis gives two products B and C both containing bromine.

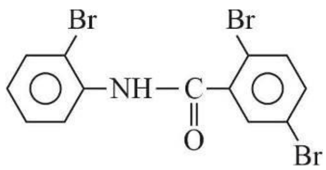
$$B \xrightarrow[H_3O^+]{NaNO_2} \xrightarrow[HBr]{CuBr} D$$

$$C \xrightarrow[CaO/\Delta]{NaOH} \text{P-dibromobenzene}$$

D gives two isomeric mono-nitro derivative and itself being isomeric with p-dibromobenzene.
Select correct options :

(A) Compound D is 

(B) Compound C is 

(C) Compound A is 

(D) Compound B is 

SECTION 3

This section contains 2 Paragraphs containing 2 Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16

A colourless solid (A) liberates a brown gas (B) on acidification; and a colour less gas (C) on treatment with NaOH, and a colourless non-reactive gas (D) on heating. If heating of the solid continued, it completely disappears.

15. A and B are, respectively:

(A) NH_4NO_3 and N_2O **(B)** NH_4NO_2 and N_2O_3

(C) NH_4NO_2 and NO_2 **(D)** N_2O_5 and HNO_2

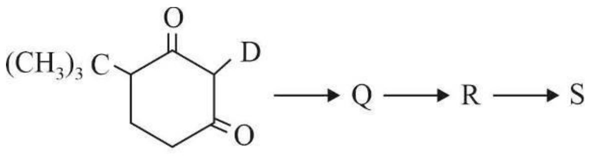
16. C and D are respectively:

(A) NO_2 and NH_3 **(B)** N_2O_3 and NH_3

(C) N_2 and NH_3 **(D)** NH_3 and N_2

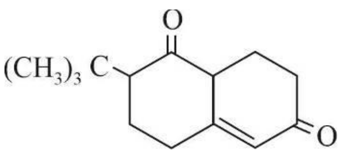
Paragraph for Q-17 to Q-18

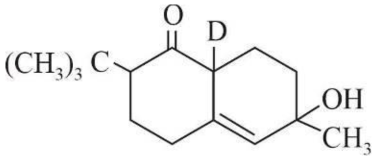
The reaction of compound P with $CH_2=CH-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$ in KOH followed by hydrolysis gives Q. The compound Q on treatment with dil. alkali and moderate heating gives compound R. The reaction of R with CH_3MgBr in $\left(C_2H_5\right)_2O$ followed by addition of H_2O produces compound S.

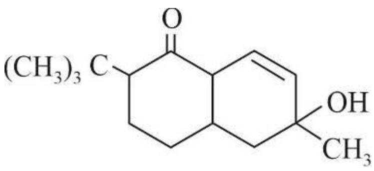


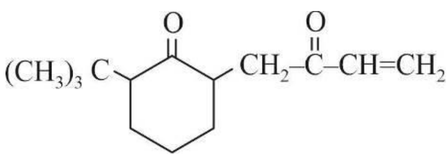
(P)

17. The product S is :

(A)


(B)


(C)


(D)


18. The reaction Q to R and R to S are :

(A) Cannizaro and electrophilic addition reaction

(B) Perkin's and condensation reaction

(C) Aldol condensation and nucleophilic addition reaction

(D) Aldol condensation and electrophilic addition reaction

SUBJECT III : MATHEMATICS 61 MARKS

SECTION 1

SINGLE CORRECT ANSWERS TYPE

This section contains 07 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let h be a twice continuously differentiable positive function on an open interval J. Let $g(x) = \ln(h(x))$ for each $x \in J$

Suppose $(h'(x))^2 > h''(x)\,h(x)$ for each $x \in J$. Then :

(A) $g(x)$ is increasing on J
(B) $g(x)$ is decreasing on J
(C) $g(x)$ is concave up on J
(D) $g(x)$ is concave down on J

2. The differential equation of family of lines situated at a constant distance α from a fixed point $(h, 0)$ is :

(A) $\left(y-\frac{dy}{dx}\right)^2=\alpha^2\left(1+\left(\frac{dy}{dx}\right)^2\right)$
(B) $\left(y-(x-h)\frac{dy}{dx}\right)^2=\alpha^2\left(1+\left(\frac{dy}{dx}\right)^2\right)$
(C) $\left(y+\frac{x\,dy}{dx}\right)^2=\alpha^2\left(1-\left(\frac{dy}{dx}\right)^2\right)$
(D) $\left(y+(x-h)\frac{dy}{dx}\right)^2=\alpha^2\left(1+\left(\frac{dy}{dx}\right)^2\right)$

3. Consider the matrices $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $B=\begin{bmatrix}a & b\\ 0 & 1\end{bmatrix}$ and let P be any orthogonal matrix and $Q = PAP^T$ and $R = P^TQ^KP$ also $S = PBP^T$ and $T = P^TS^KP$ $(a \neq 2, b \neq 2)$.

Column I	Column II
(A) If we vary K from 1 to n then the elements of the first row and first column of R will form	**(P)** G.P. with common ratio a
(B) If we vary K from 1 to n then the elements of the 2^{nd} row and 2^{nd} column of R will form	**(Q)** A. P. with common difference 2
(C) If we vary K from 1 to n then the elements of the first row and first column of T will form	**(R)** A.P. with common ratio b
(D) If we vary K from 1 to n then the first row and 2^{nd} column elements of T (where $n > 3$) will represent the sum of	**(S)** A.P. with common difference –2
	(T) A. P. with common difference a

(A) A-(Q), B-(S), C-(P), D-(P)
(B) A-(Q), B-(S), C-(R), D-(P)
(C) A-(T), B-(R), C-(P), D-(P)
(D) A-(Q), B-(S), C-(S), D-(T)

4. In a city 60% are males and 40% are females. Suppose 50% of males and 30% of females have colour blindness. One is selected at random and is found be colour blind. The probability that the selected person is male is :

(A) $\frac{10}{19}$
(B) $\frac{12}{87}$
(C) $\frac{9}{19}$
(D) $\frac{5}{7}$

5. 16 players $P_1, P_2, P_3, \ldots, P_{16}$ take part in a tennis tournament. These players are to be divided into 4 groups each comprising of 4 players. Number of ways in which 16 players can be divided into four equal groups $= K\prod_{r=1}^{8}(2r-1)$, where K equals :

(A) $\frac{35}{27}$ **(B)** $\frac{35}{24}$ **(C)** $\frac{35}{52}$ **(D)** $\frac{35}{6}$

6. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = 4$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$ and angle between $\vec{c}$ and $\vec{a}$ is $\frac{\pi}{3}$, then which of the following in incorrect

(A) Volume of parallelepiped whose adjacent edges are represented by $\vec{a}, \vec{b}$ and $\vec{c}$ is $32\sqrt{2}$

(B) Height of parallelepiped whose adjacent edges are represented by $\vec{a}, \vec{b}$ and $\vec{c}$ is $4\sqrt{\frac{2}{3}}$

(C) The volume of triangular Prism whose adjacent edges are represented by the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is $16\sqrt{2}$

(D) The volume of triangular Prism whose adjacent edges are represented by $\vec{a}, \vec{b}$ and $\vec{c}$ is $32\sqrt{2}$

7. Reflection of the line $\frac{x-1}{-1} = \frac{y-2}{3} = \frac{z-4}{1}$ in the plane $x + y + z = 7$ is

(A) $\frac{x-1}{3} = \frac{y+2}{1} = \frac{z+4}{1}$ **(B)** $\frac{x-1}{-3} = \frac{y-2}{-1} = \frac{z-4}{1}$

(C) $\frac{x-1}{-3} = \frac{y-2}{1} = \frac{z-4}{-1}$ **(D)** $\frac{x+1}{3} = \frac{y+2}{1} = \frac{z+4}{-1}$

SECTION 2

MULTIPLE CORRECT ANSWERS TYPE

This section consists of 07 Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

8. Let $f : R \to R$ is one-one, onto and differentiable function and graph of $y = f(x)$ is symmetrical about the point (4, 0), then:

(A) $f^{-1}(2020) + f^{-1}(-2020) = 8$

(B) $\int_{-2020}^{2028} f(x)\,dx = 0$

(C) If $f'(-100) > 0$, then roots of $x^2 - f'(10)x - f(10) = 0$ may be non-real

(D) If $f'(10) = 20$ then $f'(-2) = 20$

9. Let $f(x)=\begin{cases} x+1 & 0\le x\le 1 \\ 2x^2-6x+6, & 1<x\le 2 \end{cases}$ and g(t) = $\int_{t-1}^{t} f(x)\,dx$ for $t\in[1, 2]$

Which of the following hold(s) good ?

(A) $f(x)$ is continuous and differentiable in [0, 2]

(B) $g'(t)$ vanishes for t = 3/2 and 2

(C) maximum value of $g(t)$ in [1, 2] occurs at $t=\dfrac{3}{2}$

(D) minimum value of $g(t)$ in [1, 2] occurs at $t=2$

10. The area bounded by a curve, the coordinate axis & the ordinate of some point of the curve is equal to the numerical value of length of the corresponding arc of the curve. If the curve passes through the point $P(0,1)$ then :

(A) $y=\frac{1}{2}\left(e^x-e^{-x}+2\right)$ **(B)** $y=\frac{1}{2}\left(e^x+e^{-x}\right)$

(C) $y(\ln 4)=\dfrac{17}{8}$ **(D)** $y(\ln 4)=\dfrac{25}{8}$

11. $\displaystyle\sum_{r=1}^{3}\frac{1}{\tan^2\frac{r\pi}{7}}=K$ then K is co-prime to :

(A) 12 **(B)** 15 **(C)** 18 **(D)** 30

12. If $f(x)=|\log x|,$ then :

(A) Right hand derivative of f at $x=1$ is 1

(B) Left hand derivative of f at $x=1$ is 1

(C) Right hand derivative of f at $x=1$ is -1

(D) Left hand derivative of f at $x=1$ is -1

13. If $f(x)=2^{\{x\}}$ where {.} denotes the fractional part of x, then which of the following is true

(A) f is periodic **(B)** $\int_0^1 2^{\{x\}}dx=\dfrac{1}{\log 2}$

(C) $\int_0^1 2^{\{x\}}dx=\log_2 e$ **(D)** $\int_0^{100} 2^{\{x\}}dx=100\log_2 e$

14. Consider the function $f:R\to R$ defined as $f(x)=x+\sin x$. Which of the following is/are the correct statement(s)?

(A) The function is strictly increasing at every point on R except at 'x' equal to an odd integral multiple of π where the derivative of $f(x)$ is zero and where the function f is not strictly increasing

(B) The function is bounded in every bounded interval but unbounded on whole real line

(C) The graph of the function $y=f(x)$ lies in the first and third quadrants only

(D) The graph of the function $y=f(x)$ cuts the line $y=x$ at infinitely many points

SECTION 3

This section contains 2 Paragraphs containing 2 Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16

Let A, B, C be the vertices of a triangle ABC in which B is taken as origin of reference and position vectors of A and C are $\vec{a}$ and $\vec{c}$ respectively. A line AR parallel to BC is drawn from A. PR (P is mid-point of AB) meets AC at Q and area of triangle ACR is 2 times area of triangle ABC.

15. Position vector of R in terms of $\vec{a}$ and $\vec{c}$ can be :

(A) $\vec{a}+2\vec{c}$ **(B)** $\vec{a}+3\vec{c}$ **(C)** $\vec{a}+\vec{c}$ **(D)** $\vec{a}+4\vec{c}$

16. Position vector of Q can be :

(A) $\dfrac{2\vec{a}+3\vec{c}}{5}$ **(B)** $\dfrac{3\vec{a}+2\vec{c}}{5}$ **(C)** $\dfrac{\vec{a}+\vec{c}}{2}$ **(D)** $\dfrac{\vec{a}+3\vec{c}}{4}$

Paragraph for Q-17 to Q-18

Suppose two quadratic equations $a_1x^2+b_1x+c_1=0$ and $a_2x^2+b_2x+c_2=0$ have a common root α, then

$a_1\alpha^2+b_1\alpha+c_1=0$ ...(1) and $a_2\alpha^2+b_2\alpha+c_2=0$ ...(2)

Eliminating α using cross-multiplication method gives us the condition for a common root. Solving two equations simultaneously, the common root can be obtained.

Now, consider three quadratic equations, $x^2-2p_rx+r=0;\ r=1,2,3$ given that each pair has exactly one root common. If common root between 1st and 2nd is α, between 2nd and 3rd is γ and 1st and 3rd be β, then:

17. $\dfrac{\gamma}{\beta}$ is equal to :

(A) 2 **(B)** 3 **(C)** 1 **(D)** 1/2

18. $\dfrac{\gamma}{\alpha}$ is equal to :

(A) 2 **(B)** 3 **(C)** 1/2 **(D)** 1/3

SPACE FOR ROUGH WORK

End of JEE Advanced-10 | PAPER-2 | JEE-2022